



Chapter 3

System Architecture and Technology Stack

3.1 System Architecture Overview

The Integrated Housing Services and Monitoring System (IHSMS) is designed as a centralized web-based information system that supports the operational processes of the Talisay City Housing Authority. The system integrates key housing governance functions such as Informal Settler Family monitoring, housing application management, document tracking, housing project and unit monitoring, case management and operational reporting.

The system is developed to address the fragmented and manual processes previously used by the Housing Authority in managing housing-related records. By consolidating these processes into a single digital platform, the system enables authorized personnel to record, track and monitor housing data more efficiently.

To support organized system processing, the platform follows a **three-tier client-server architecture** consisting of the Presentation Layer, Application Layer and Data Layer. These layers work together to allow authorized staff to interact with the system through web interfaces while application logic processes requests and stores data in a centralized database.

3.2 Architecture Model

The system adopts a **client-server architecture model** in which authorized personnel of the Talisay City Housing Authority access the system through a web browser, while system operations are processed by the application server and stored in a centralized database. This architecture is composed of three major layers: the **Presentation Layer (User Interface Layer)**, the **Application Layer (Business Logic Layer)** and the **Data Layer (Database Layer)**. Through this layered structure, user requests are first handled by the interface, processed by the application layer, and then stored or retrieved from the database layer.

The system architecture follows the structure shown below:

Authorized Housing Authority Staff



Presentation Layer



(Web-Based User Interface)



Application Layer

(System Modules and Processing Logic)



Data Layer

(Centralized Housing Database)

This layered architecture allows the system to organize housing data efficiently while ensuring secure, controlled, and structured access to system functions.

3.3 Presentation Layer

The Presentation Layer represents the user interface of the system and provides web-based interfaces that allow authorized personnel of the Talisay City Housing Authority to interact with the platform. Through this layer, administrative and technical staff can encode **housing-related data**, **manage records**, **monitor housing applications**, **track document submissions**, **review housing project information**, **record cases or complaints**, and **generate reports**.

The presentation layer includes major interface screens such as the **login page**, **system dashboard**, **Informal Settler Family management interface**, **housing application management interface**, **document tracking interface**, **housing project monitoring interface**, **case management interface**, and the **analytics and reporting dashboard**. These interfaces provide a structured environment that allows staff to efficiently access system functions and manage housing records across the city.

3.4 Application Layer

The **Application Layer** contains the business logic and processing components of the system. It manages system operations and ensures that all activities follow the defined workflow of the Talisay City Housing Authority. This layer processes user requests from the presentation layer, validates data, updates records, and communicates with the database layer.

The application layer integrates the core modules of the system. The Informal Settler Family Monitoring Module maintains records of Informal Settler Families (ISFs) in Talisay City, allowing staff to record household information and monitor ISF distribution across barangays.



Since housing applicants are typically selected from ISF records, this module serves as the primary data source for housing program management.

The Housing Application Management Module records housing applications submitted by Informal Settler Families and allows staff to track the progress of applications from submission to final status. The Document Management and Tracking Module monitors the completeness of application requirements and allows staff to verify whether required documents have been submitted, verified, or remain incomplete.

The Housing Project and Unit Monitoring Module records housing project information and monitors the availability of housing units, enabling staff to assign approved applicants to available units. The Case Management Module records housing-related concerns or complaints and tracks their progress through stages such as received, under review, in progress, resolved, and closed.

In addition, the Analytics and Reporting Dashboard generates summarized reports and visual dashboards that display ISF statistics, housing application status, housing unit availability, and case records to support operational monitoring and decision-making. The application layer also implements Role-Based Access Control to ensure that system access and permissions are restricted according to the roles of authorized personnel.

3.5 Data Layer

The **Data Layer** is responsible for managing the centralized storage of all information processed by the Integrated Housing Services and Monitoring System. This layer uses a relational database management system to store records generated by the different system modules. The database supports the storage, retrieval, and updating of housing-related information and ensures that records are consistently organized and accessible for system monitoring and reporting purposes.

The database maintains several key data entities required for system operations. These include user accounts and roles that manage system access and permissions for authorized personnel. It also stores Informal Settler Family records that contain demographic and household information used as the foundation for housing program monitoring. In addition, the system records housing application information submitted by Informal Settler Families, including the status of applications and associated program details.

The database also manages records related to application document submissions, housing project information, and housing unit availability. These records support the monitoring of housing



projects and the assignment of housing units to qualified applicants. Furthermore, the system stores case records related to housing concerns or complaints, allowing staff to track the progress and resolution of reported cases.

To support accountability and traceability of system activities, the database also stores audit trail records that log significant actions performed within the system. In addition, summarized operational data used by the analytics and reporting dashboard is maintained to support the generation of reports and statistical summaries.

By maintaining a centralized database, the system improves data accessibility and supports more efficient monitoring and management of housing programs implemented by the Talisay City Housing Authority.

3.6 System Workflow

The Integrated Housing Services and Monitoring System follows a structured workflow that connects the different modules of the platform. The process begins when staff register Informal Settler Family records in the system, which serve as the base data for identifying potential housing program applicants.

When an Informal Settler Family applies for a housing program, staff create a housing application record in the application management module. The document management module is then used to verify whether the applicant has submitted all required documents. Once the documents are verified, the application proceeds to the evaluation stage. Qualified applicants may then be approved and assigned to available housing units recorded in the housing project monitoring module.

The system also allows staff to record housing-related concerns or complaints through the case management module. All data generated from these processes are automatically summarized in the analytics and reporting dashboard, enabling management to monitor housing operations and program implementation.

3.7 Technology Stack

The Integrated Housing Services and Monitoring System is implemented using modern web development technologies that support efficient development, reliability, and maintainability. The system uses a combination of front-end technologies, back-end frameworks, database management systems, and development tools to support the functionality of the platform.



Front-End Technologies

The presentation layer of the system is developed using several web technologies that support the structure, design, and interactivity of the user interface. **HTML5** is used to structure the content and layout of the web pages, providing the foundation for displaying system components and user interface elements. **CSS3** and **SCSS** are utilized to design the visual layout and ensure responsive interface behavior across different screen sizes. In addition, **JavaScript** is used to implement interactive components and dynamic behavior within the system. These technologies allow the system to provide a structured, responsive and user-friendly interface for authorized personnel of the Talisay City Housing Authority.

Back-End Technologies

The application logic of the system is implemented using server-side technologies that support system processing and data handling. **Python 3** is used as the primary programming language for server-side development due to its efficiency and strong support for web application development. The **Django Web Framework** is utilized to develop the system's core functionality, including user authentication, database interaction, and integration of system modules. In addition, the **Django REST Framework** is used to handle application programming interface (API) communication and structured data exchange within the system. During development, **Postman** is used to test and verify API endpoints to ensure proper communication between system components. These technologies support efficient development and secure processing of system operations.

Database System

The system uses a relational database management system to store and manage housing-related records. **PostgreSQL** is used as the primary database management system for storing structured data such as Informal Settler Family records, housing applications, housing project information, and case records. To support database management during development, **Beekeeper Studio** is used as a database administration tool that allows developers to access, manage, and organize database tables efficiently.

Version Control

Version control tools are used to manage source code and track system development changes throughout the project lifecycle. **Git** is used as the primary version control system to monitor modifications in the system's source code. **GitHub** serves as the repository platform where



project files and code versions are stored and managed collaboratively. Additionally, **GitKraken** is used as a graphical user interface for managing Git repositories and performing version control operations more efficiently.

Development Environments

The system is developed using integrated development environments that support efficient coding, debugging, and project management. **Visual Studio Code** is used primarily for front-end development and general web application coding. **PyCharm** is used for Python and Django development due to its specialized support for Python programming and web frameworks. These development environments provide debugging tools, code organization features, and development support that facilitate the efficient implementation of the Integrated Housing Services and Monitoring System.

3.8 Architectural Benefits

The architecture selected for the Integrated Housing Services and Monitoring System provides several advantages that support the operational needs of the Talisay City Housing Authority. By adopting a layered system architecture, the platform separates system components into the user interface, application processing, and data management layers, improving system organization and making the system easier to maintain and update.

The use of a centralized database also improves the accessibility and consistency of housing-related records. By storing Informal Settler Family data, housing applications, housing project information, and case records in a single system, authorized personnel can retrieve and manage information more efficiently compared to the previously used manual and fragmented processes. This centralized approach supports more effective monitoring of housing programs and improves the reliability of operational reports.

The implementation of role-based access control further enhances system security by ensuring that only authorized personnel can access specific system functions based on their assigned roles. In addition, the modular structure of the system allows future improvements or additional features to be integrated without significantly affecting existing system operations. Through this architecture, the system provides a structured, secure, and scalable platform that supports the housing management operations of the Talisay City Housing Authority.